

CS 312 - Project Phase – 2 Report

Group Details:

Student 1 Name	Tyler Vander Mooren
Student 1 Email	Tjv55@nau.edu

Phase-2 Details:

Project Title	Employee Database Management System (EDMS)
Phase - 2 Components	• Implement role-based access logic. • Complete CRUD functionality (frontend + backend). • Build employee list and profile views. • Add search, filter, and sort functionality for admins. • Finalize styling, test UX, and write project documentation.

Task Status:

Task No	Task Details	Assigned To	Task Status
1	Implement role-based middleware and frontend access control • Write Express middleware to check for admin vs employee role before accessing routes like POST /add, DELETE /employee/:id. • In React, restrict admin-only features from being shown to regular users.	Tyler Vander Mooren	Completed

2	Create and test full CRUD routes for employee records - Backend: Implement routes like GET /employees, POST /employee, PUT /employee/:id, DELETE /employee/:id. - Frontend: Build forms/components to call these routes based on user role.	Tyler Vander Mooren	Completed
3	Develop Admin dashboard with full employee table - Display employee records in a table. - Include Edit and Delete buttons with confirmation prompts. - Fetch employee data from backend and handle updates live.	Tyler Vander Mooren	Completed
4	Build Employee Profile View/Edit page - For employees, display their profile data on login. - Provide a form to update their name, email, position, etc. - Prevent access to any other user's data from the frontend.	Tyler Vander Mooren	Modified
5	Implement search, filter, and sort functionality for employee records (admin only) - Add a search input to filter employees by name, ID, or position. - Implement sorting by name, join date, or position using dropdowns or column headers.	Tyler Vander Mooren	Completed.
6	Finalize frontend styling and write project documentation	Tyler Vander Mooren	Completed/backlog

	• Apply consistent CSS or Bootstrap styling for professional UI. • Write README.md including project setup, features, technologies, and usage instructions.		
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New Tasks or Modified Tasks (Optional):

Task Type	Task Details	Assigned To	Task Status
Modified	• Provide a form to update their name, email, position and password. Only does name, email, position right now	Tyler Vander Mooren	Backlog
Backlog	Write README.md including project setup, features, technologies, and usage instructions.	Tyler Vander Mooren	Backlog

Challenges and Solutions Implementation:

List at least 4 challenges faced by your team during phase-2 development and elaborate on the solutions you have implemented. Feel free to add more rows to the following table.

Details of the Challenge	Solution Implemented
Integrating role-based access control to restrict signup functionality to admin users only, while managing sessions securely.	Implemented middleware checks (isLoggedIn and isAdmin) on backend routes to enforce role-based access and used secure sessions with proper cookie handling. Also updated frontend routing to reflect permissions.
Syncing frontend API calls with backend endpoints caused 404 errors due to mismatched route URLs or ports.	Standardized all API endpoints to use a common base URL with withCredentials enabled for cookies. Reviewed and aligned frontend route calls with backend routes to fix errors.

Managing employee data schema inconsistencies between frontend forms and backend PostgreSQL tables led to insertion errors.	Conducted a detailed schema review and refactored backend queries and frontend form fields to align with the database schema, removing unnecessary fields like phone and department to prevent errors.
Mistook Error messages in admin dashboard as code not working.	It simply was an error saying that this email was already used in creation. The solution is to fix it in the final phase to show correct error messages for the admin.

Phase-2 Developed Screens:

Welcome to the EDMS Portal

Employee Database Management System developed by Tyler Vander Mooren

Sign In

Access your dashboard securely

Employee Sign In

Admin Dashboard - Employee Management

Search employees...

Add User

Name	Position	Email	Actions	
Testedit Filleredit	Testeredit	pleasetest@yahoo.com	Edit	Delete
Edit Profile	tester	editprofile@test.com	Edit	Delete
searchbartest test	tester	searchbar@test.com	Edit	Delete
delete test	tester	tester@delete.com	Edit	Delete

My Profile

Name

Email

editprofile@test.com

Position

tester

Update